

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks available |          |          |           |          |          |
|----------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |     |       | <b>Another alternative for full marks:</b><br>$KE = \frac{1}{2} m \overline{c^2}$ (1)<br>Substituting from $pV = \frac{1}{3} Nm \overline{c^2}$ into above e.g. $KE = \frac{1}{2} \frac{3pV}{N}$ (1)<br>Substituting $pV = NkT$ into above e.g. $KE = \frac{1}{2} \frac{3NkT}{N}$ (1)<br>Hence $KE = \frac{3}{2} kT$ (1)<br><br><b>Alternative for a maximum of 3 marks:</b><br>Internal energy or $U = \frac{3}{2} nRT$ (1)<br>$U = \frac{3}{2} NkT$ (1)<br>This is for $N$ molecules, hence $KE = \frac{3}{2} kT$ (1) |                 |          |          |           |          |          |
|          |     | (ii)  | $\frac{3}{2} \times 1.38 \times 10^{-23} \times 293 = 6.07 \times 10^{-21}$ [J]                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 | 1        |          | 1         | 1        |          |
|          |     | (iii) | $4.75 \times 10^{22}$ <b>ecf</b> $\times 6.07 \times 10^{-21}$ <b>ecf</b> = 288 [J]                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 | 1        |          | 1         | 1        |          |
|          | (c) | (i)   | Ideal gas – kinetic energy only or no PE (1)<br>Liquid – kinetic energy and potential energy (1)<br>Link PE to intermolecular forces or separation of molecules or intermolecular force / separation mentioned for liquid (1)                                                                                                                                                                                                                                                                                           | 3               |          |          | 3         |          |          |
|          |     | (ii)  | At absolute zero or 0 [K] or -273 [°C]<br>Do not accept 0 °C OR -273 K (treat as neutral, don't penalise – they may have said something else that is correct e.g. absolute zero)                                                                                                                                                                                                                                                                                                                                        | 1               |          |          | 1         |          |          |
|          |     |       | <b>Question 2 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>9</b>        | <b>5</b> | <b>0</b> | <b>14</b> | <b>8</b> | <b>0</b> |